

MODULE *messages*

Message constructors for the *Zcash P2P* wire protocol (*ZIP-0204*).

Each operator builds a record with a header (magic number, command name) and a *payload* whose fields mirror the on-wire format. Fields that are irrelevant to the properties being verified are set to constant values to keep the *TLC* state space manageable.

EXTENDS *Naturals*, *FiniteSets*

$MakeVersion(from, to, blocks) \triangleq [$
 $header \mapsto [magic \mapsto 619259748, command \mapsto "version"],$
 $payload \mapsto [$
 $version \mapsto 170100,$
 $services \mapsto 0,$
 $addr_recv \mapsto to,$
 $addr_from \mapsto from,$
 $nonce \mapsto 0,$
 $user_agent \mapsto "",$
 $start_height \mapsto Cardinality(blocks),$
 $relay \mapsto FALSE$
 $]$
 $]$

$MakeVerack \triangleq [$
 $header \mapsto [magic \mapsto 619259748, command \mapsto "verack"]$
 $]$

$MakePing \triangleq [$
 $header \mapsto [magic \mapsto 619259748, command \mapsto "ping"],$
 $payload \mapsto [nonce \mapsto 0]$
 $]$

$MakePong \triangleq [$
 $header \mapsto [magic \mapsto 619259748, command \mapsto "pong"],$
 $payload \mapsto [nonce \mapsto 0]$
 $]$

$MakeInv(blocks) \triangleq [$
 $header \mapsto [magic \mapsto 619259748, command \mapsto "inv"],$
 $payload \mapsto [$
 $count \mapsto 1,$
 $inventory \mapsto \langle [type \mapsto "MSG_BLOCK", hash \mapsto Cardinality(blocks)] \rangle$
 $]$
 $]$

$MakeGetHeaders(blocks) \triangleq [$
 $header \mapsto [magic \mapsto 619259748, command \mapsto "getheaders"],$
 $payload \mapsto [$
 $version \mapsto 170100,$
 $]$
 $]$

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    hashCount      ↦ 1,
    blockLocatorHashes ↦ ⟨Cardinality(blocks)⟩,
    hashStop       ↦ 0
  ]
]

MakeHeaders(blocks) ≜ [
  header ↦ [magic ↦ 619259748, command ↦ "headers"],
  payload ↦ [
    count ↦ 1,
    headers ↦ ⟨[version ↦ 170140, prev_block ↦ Cardinality(blocks), merkle_root ↦ 0, bits ↦ 0, nonce ↦ 0]⟩
  ]
]

MakeGetData(blocks) ≜ [
  header ↦ [magic ↦ 619259748, command ↦ "getdata"],
  payload ↦ [
    count ↦ 1,
    inventory ↦ ⟨[type ↦ "MSG_BLOCK", hash ↦ Cardinality(blocks)]⟩
  ]
]

MakeReject(rejected_command) ≜ [
  header ↦ [magic ↦ 619259748, command ↦ "reject"],
  payload ↦ [message ↦ rejected_command]
]

MakeBlock(blocks) ≜ [
  header ↦ [magic ↦ 619259748, command ↦ "block"],
  payload ↦ [
    version ↦ 70015,
    prev_block ↦ Cardinality(blocks),
    merkle_root ↦ 0,
    bits ↦ 0,
    nonce ↦ 0,
    transactions ↦ ⟨⟩
  ]
]

```
